INCH-POUND
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SUPERSEDING
(See 6.5)

DETAIL SPECIFICATION

STEEL FORGINGS, CARBON AND ALLOY HEAT-TREATED

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers carbon and alloy heat-treated steel forgings for general shipboard application.

1.2 Classification. The carbon and alloy steel forgings are of the following types and classes, as specified (see 6.2).

1.2.1 Classes. Steel forgings are of the classes specified in table I:

TABLE I. Steel forging classes.

Class	Yield strength (minimum) (lb/in^2)
A	140,000
B	120,000
C	100,000
D	80,000
E	65,000
F	45,000
G	40,000
H	30,000

1.2.2 Types. Steel forgings are of the types specified in table II:

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AMSC N/A AREA FORG

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TABLE II. Steel forging types.

Type	Alloy
I	Ni Cr Mo
II	Cr Mo
III	Ni
IV	Plain carbon (medium)
V	Plain carbon (low)

1.2.3 Part or identifying number (PIN). Part numbers for specific hexagonal and round bars covered in this specification will be formulated as described in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) TECHNICAL PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM A751 - Standard Test Methods and Practices for Chemical Analysis of Steel Products

ASTM A788/A788M - Standard Specification for Steel Forgings, General Requirements

(Copies of these documents are available online at www.astm.org.)

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SAE INTERNATIONAL

SAE-AMS-2806 - Identification Bars, Wire, Mechanical Tubing, and Extrusions, Carbon and Alloy Steels and Corrosion and Heat-Resistant Steels and Alloys

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material from which the forgings are made shall be as specified in 3.1.1 through 3.1.4.

3.1.1 Melting and casting. The material from which the forgings are made shall be manufactured by electric furnace, basic oxygen, or vacuum induction melting process. Unless otherwise specified (see 6.2), the material shall be continuous cast or cast in metal molds. The primary melting may incorporate separate degassing or refining and may be followed by secondary melting using electroslag remelting (ESR) or vacuum arc remelting (VAR), as specified (see 4.9 and 6.2).

3.1.2 Discard. Sufficient discard shall be taken from each ingot (produced of a metal mold or continuous caster) to ensure freedom from piping and to prevent undue segregation (see 4.4.1, 4.5, and 4.5.4.1).

3.1.3 Ladle (heat) chemical composition. The ladle (heat) chemical composition shall be in accordance with table III (see 4.5.1, 4.5.2, and 4.9).

TABLE III. Chemical composition (ladle analysis and check analysis). 1/, 2/

Type	Carbon	Manganese	Phosphorous	Sulfur	Silicon	Copper	Nickel	Chromium	Molybdenum
I	0.44	0.90	0.025	0.025	0.20/0.35	-	1.65/2.00	0.70/0.95	0.20/0.30
II	0.44	1.10	0.025	0.025	0.20/0.35	-	-	0.80/1.10	0.15/0.25
III	0.40	0.85	0.025	0.025	0.20/0.35	-	3.25/3.75	-	-
IV 3/	0.44	0.90	0.04	0.04	0.10/0.30	-	-	-	-
V 3/	0.30	0.90	0.04	0.04	0.10/0.30	0.25	0.25	-	-

NOTES:

1/ Maximum weight percent unless range is shown.

2/ Intentional alloying additions shall not be permitted for elements with an unidentified limit.

3/ Types IV and V are designed to be plain carbon steels; intentional alloying additions shall not be permitted.

3.1.4 Recycled, recovered, or environmentally preferable materials. Recycled (see 6.6.5), recovered (see 6.6.4), or environmentally preferable (see 6.6.2) materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. When specified (see 6.2), virgin raw materials shall be used (see 4.9).

3.2 Forging process. Unless otherwise specified (see 6.2), the forging process requirements shall be as specified in 3.2.1 through 3.2.4.

3.2.1 Hot rolling. Hot rolling is permitted only for hexagonal and round bars conforming to class B, types I and II as described in 6.7, where the A dimension is less than or equal to 2.000 inches (see 4.9).

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3.2.2 Reduction. The total reduction of the cross-sectional area from ingot or billet to forging shall be not less than 3 to 1. Palms, flanges, and other enlargements on forgings need not be reduced to the ratio of 3 to 1 but shall be reduced in a ratio of not less than 1.7 to 1. If bored ingots or billets are used, the wall of the ingot shall be reduced to at least ½ of its original thickness, or the total reduction of area shall be at least 3 to 1. Where an upsetting operation is employed, or the forging expanded on a mandrel, the metal shall be worked to an extent of not less than indicated within this paragraph, but no fixed ratio between the cross-sectional areas of the ingot or billet and of the forging is required (see 4.4.2 and 4.9).

3.2.3 Bored ingots. When the ingots are bored, the centerline of the bored hole shall include the centerline of the ingot (see 4.4.3 and 4.9).

3.2.4 Flame cutting. Metal removal by flame cutting or other thermal means shall not be permitted after the forging has received its final thermal treatment and final nondestructive tests (see 4.9).

3.3 Forging (check) chemical composition. The forging (check) chemical composition of the steel forgings shall be as specified in table III subject to the tolerances specified in ASTM A788/A788M (see 4.5, 4.5.1, and 4.9).

3.4 Mechanical properties. The mechanical property requirements shall be as specified in 3.4.1 and 3.4.2.

3.4.1 Tensile properties. Carbon and alloy steel forging tensile properties shall be as specified in table IV (see 4.5, 4.5.2, and 4.9).

TABLE IV. Tensile properties.

Class	Type	Section thickness	Yield strength (0.2 percent offset)	Ultimate tensile strength		Elongation in 2 inches gauge		Reduction of area	
						Longitudinal 1/	Transverse 2/, 3/	Longitudinal 1/	Transverse 2/, 3/
		Minimum	Minimum	Minimum	Maximum	Minimum	Minimum	Minimum	Minimum
		(inches)	(lb/in^2)	(lb/in^2)	(lb/in^2)	(percent)	(percent)	(percent)	(percent)
A	I	10	140,000	165,000	-	10	8	35	20
B	I or II	10	120,000	140,000	-	14	10	35	25
C	I or II	10	100,000	120,000	-	16	12	40	30
D	I, II, or III	10	80,000	100,000	-	18	14	45	30
E	I, II, or III	10	65,000	95,000	-	20	17	45	30
F	II, III, or IV	-	45,000	-	90,000	22	19	45	30
G	IV	-	40,000	-	90,000	22	19	45	30
H	V	-	30,000	-	90,000	30	25	45	40

NOTES:

1/ The term longitudinal for purposes of this specification means that the major axis (length) of the test specimen shall be parallel to the direction that the steel is most drawn (parallel with metal flowlines).

2/ The term transverse means the major axis (length) of the test specimen shall be perpendicular to the direction that the steel is most drawn (normal to metal flowlines and parallel to the plane of the major surface).

3/ Transverse tensile requirement shall be applicable only when specified (see 6.2).

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3.4.2 Impact requirements. When specified (see 6.2), steel forgings shall meet the impact property requirements of table V (see 4.5, 4.5.3, and 4.9).

TABLE V. Impact properties.

Class	Type	Test temperature	Charpy "V" notch values average of three tests 1/ (minimum)
		°F±3°	(ft-lbs)
A	I	10	15
B	I or II	10	15
C	I or II	10	30
D	I, II, or III	10	30
E	I, II, or III	10	30
F	II or III	10	30
G	IV	10	15
H	V	10	15

NOTE:

1/ No single test shall fall below the specified value by more than 5 foot-pounds (ft-lbs).

3.5 Heat treatment. The heat treatment shall be as specified in 3.5.1 through 3.5.3 (see 4.9).

3.5.1 Heat treatment procedures. Each forging shall receive a detailed heat treatment in accordance with a written procedure that will produce forgings to meet the requirements of this specification.

3.5.2 Class restrictions. In order to develop the required properties, all classes, except class H, shall be either normalized and tempered or quenched and tempered in their final heat-treated condition. Class H shall be either annealed, normalized, or normalized and tempered in its final heat-treated condition. Tempering temperature shall not be below 950 degrees Fahrenheit (°F) for class A, 1000 °F for class B, and 1050 °F for classes C through H.

3.5.3 Stress relieving. Unless otherwise specified (see 6.2), a stress relief treatment may be performed on forgings after final machining, in which case the stress relief temperature shall be at least 50 °F below the tempering temperature applied to the forging. For stress relief after weld repairs, see 3.8.4.

3.6 Dimensions. Forgings shall conform to the dimensions and tolerances specified (see 6.2). The forgings shall have sufficient stock to permit finishing (see 4.4.4).

3.7 Identification marking. Unless otherwise specified (see 6.2), all forgings shall be legibly and indelibly marked in accordance with SAE-AMS-2806 and the following (see 4.4.5):

a. Include a forging number that identifies the manufacturer, heat, heat treatment lot, specification, class, dimensions, and type of forging.

b. Unless otherwise specified (see 6.2), each forging weighing greater than 250 pounds shall be marked with the contractor's name or trademark, the specification number, class, type, dimensions, drawing or die number, part number (if applicable), and heat or forging number.

c. When specified (see 6.2), the markings shall be indentation stamped.

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3.8 Soundness. Soundness shall be as specified in 3.8.1 and 3.8.2 (see 4.5.4).

3.8.1 Surface soundness. Each forging shall be free from defects that may detrimentally affect their suitability for the service intended as determined by the applicable nondestructive tests (see 4.5.4.1, 4.5.4.2, and 4.9).

3.8.2 Internal soundness. Internal soundness of each forging shall meet the acceptance criteria in 4.5.4.3.2 when specified (see 4.5.4.3, 4.9, and 6.2).

3.8.3 Repair of surface defects. Defects shall be removed, to the extent necessary to meet the section 4 criteria for the applicable nondestructive methods, by grinding, chipping, or other mechanical means provided the width of the involved area is three times its depth and gradually tapers into the defects and the design dimensions shall be maintained in accordance with 3.6. Heat shall not be applied to remove defects after heat treatment or stress relief (see 4.9).

3.8.4 Repair by welding. Forgings shall be repaired by welding only with specific case by case approval, as specified (see 6.2). Requirements for repair by welding shall be as specified (see 4.9 and 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.2).

4.2 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.5 as summarized in table VI.

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TABLE VI. Summary of conformance inspections.

Characteristic	Conformance	Requirement	Verification
Material	X	3.1	-
Melting and casting	X	3.1.1	4.9
Discard	X	3.1.2	4.4.1
Ladle (heat) chemical composition	X	3.1.3	4.5.1, 4.9
Virgin raw materials	When specified (see 6.2)	3.1.4	4.9
Forging process	X	3.2	-
Hot rolling	X	3.2.1	4.9
Reduction	X	3.2.2	4.4.2, 4.9
Bored ingots	X	3.2.3	4.4.3, 4.9
Flame cutting	X	3.2.4	4.9
Forging (check) chemical composition	X	3.3	4.5, 4.5.1, 4.9
Mechanical properties	X	3.4	-
Tensile properties	X	3.4.1	4.5, 4.5.2, 4.9
Impact requirements	When specified (see 6.2)	3.4.2	4.5, 4.5.3, 4.9
Heat treatment	X	3.5, 3.5.1, 3.5.2, 3.5.3	4.9
Dimensions	X	3.6	4.4.4
Identification marking	X	3.7	4.4.5
Soundness	X	3.8	4.5.4
Surface soundness	X	3.8.1	4.5.4.1, 4.5.4.2, 4.9
Internal soundness	When specified (see 6.2)	3.8.2	4.5.4.3, 4.9
Repair of surface defects	X	3.8.3	4.9
Repair by welding	X	3.8.4	4.9

4.3 Sampling. Sampling shall be as specified in 4.3.1 through 4.3.3.5.

4.3.1 Lot. A lot shall be as specified in 4.3.1.1 and 4.3.1.3.

4.3.1.1 Chemical analysis. For purposes of sampling for chemical or spectrographic analysis, a lot shall consist of all forgings from one heat or melt (see 3.3).

4.3.1.2 Tension and impact testing. For purposes of tension and impact testing, a lot shall consist of the following:

a. All forgings of the same type and size, with an as-heat-treated weight of less than 1,000 pounds produced from one heat, and heat treated in the same furnace charge or by a continuous process under the same conditions of treating over a period of 4 hours.

b. Each forging with an as-heat-treated weight of 1,000 pounds or more.

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4.3.1.3 Lot for dimensional and surface examination. A lot shall consist of all bars of the same type and size offered for inspection at one time.

4.3.2 Number of test specimens. The number of test specimens shall be as specified in 4.3.2.1 through 4.3.2.4.

4.3.2.1 Ladle (heat) chemical analysis. One test sample shall be taken during the pouring of the heat at a time that best represents the composition of the cast.

4.3.2.2 Forging (check) chemical analysis. One chemical check analysis specimen shall be taken from each lot.

4.3.2.3 Tension testing. The number of test specimens for tension testing shall be as specified in 4.3.2.3.1 through 4.3.2.3.3.

4.3.2.3.1 Forgings weighing less than 1,000 pounds. One tension test specimen shall be taken from each lot.

4.3.2.3.2 Forgings weighing 1,000 pounds or more and 8 feet and longer. One tension test specimen shall be taken from each specified location of each forging.

4.3.2.3.3 Forgings weighing 1,000 pounds or more and less than 8 feet long. Each forging shall have one tension test specimen taken from the specified location in 4.3.3.

4.3.2.4 Impact testing. One set of three impact specimens shall be taken from each lot.

4.3.3 Location of test specimens. The location of test specimens shall be as specified in 4.3.3.1 through 4.3.3.5.

4.3.3.1 Prolongations. Prolongations shall be provided on forgings for taking tests, except that for small forgings that are heat treated in groups, a forging may be destroyed in lieu of providing prolongations.

4.3.3.2 Ladle (heat) chemical analysis. Samples shall be taken from the molten stream while pouring the ladle contents into the mold.

4.3.3.3 Forging (check) chemical analysis. Samples shall be taken from broken tension test specimens.

4.3.3.4 Tension testing. The specimens for tension testing shall be taken at mid-radius of rounds; mid-wall thickness of bored forgings; or mid-thickness, quarter width of other forgings, and as specified in 4.3.3.4.1 through 4.3.3.4.3.

4.3.3.4.1 Tension testing of forgings weighing less than 1,000 pounds. The specimen shall be taken from the forging or prolong nearest the top of the as-cast ingot for forgings forged directly from ingot. For forgings forged directly from billet, the specimen shall be selected from a prolongation or a randomly selected forging.

4.3.3.4.2 Tension testing of forging weighing 1,000 pounds or more and 8 feet and longer. The specimens shall be taken from each end of the forging.

4.3.3.4.3 Tension testing of forging weighing 1,000 pounds or more and less than 8 feet long. The specimen shall be taken from the top (with respect to the as-cast ingot) for forgings forged directly from ingot. For forgings forged directly from billet, the specimen shall be selected from a prolongation or a randomly selected forging.

4.3.3.5 Impact testing. Specimens for impact testing shall be taken from the top prolongation of the forgings with respect to the as-cast ingot for forgings forged directly from ingot. For forgings forged directly from billet, the specimen shall be selected from a prolongation or a randomly selected forging. The specimens shall be taken at least 1 inch below the forged surface. The major axis (length) of the specimen shall be taken in the direction that the steel is most drawn. The major axis (length) of the notch shall be perpendicular to the nearest forged surface.

4.3.4 Defective specimens. If any test specimen reveals obvious defective machining or obvious lack of continuity of metal, it may be discarded and another selected.

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4.4 Examinations. Examinations shall be as specified in 4.4.1 through 4.4.5.

4.4.1 Discard. The amount of discard taken from each forging shall be verified by visual inspection (see 4.5.4.1) and any additional means as determined by the manufacturer (see 3.1.2).

4.4.2 Reduction. The reduction ratio for each forging shall be verified by any means as determined by the manufacturer (see 3.2.2).

4.4.3 Bored ingots. The location of the boring centerline in each bored ingot shall be verified by any means as determined by the manufacturer (see 3.2.3).

4.4.4 Dimensional examinations. Each forging shall be examined for conformance to the specified dimensions and tolerances by any means capable of equivalent or higher precision than that of the required dimensions and tolerances. Dimensions shall be verified on each completed forging (i.e., after final heat treatment, final stress relief, and final machining operations) (see 3.6).

4.4.5 Identification marking. Verification of identification markings shall be by visual inspection or other means as determined by the manufacturer (see 3.7).

4.5 Tests. Samples selected in accordance with 4.3, as applicable, shall be subject to the tests for conformance inspections as designated in table VI and as specified in 4.5.1 through 4.5.4.3.2.

4.5.1 Chemical analysis. Chemical or spectrographic analysis shall be in accordance with ASTM A751 (see 3.1.3 and 3.3).

4.5.2 Tension test. Standard round tension test specimens shall be tested in accordance with ASTM A370. Tension tests shall be performed after the final heat treatment (see 3.4.1).

4.5.3 Charpy “V” notch impact test. When impact properties are required (see 3.4.2), the Charpy “V” notch test specimen shall be tested in accordance with ASTM A370 at the temperature specified in table V. Impact tests shall be performed after the final heat treatment (see 3.4.2).

4.5.4 Nondestructive tests. Nondestructive tests shall be as specified in 4.5.4.1 through 4.5.4.3.2.

4.5.4.1 Visual testing (VT) prior to packaging. VT shall be performed as specified in 4.5.4.1.1 and 4.5.4.1.2 (see 3.8.1).

4.5.4.1.1 Method. Each forging shall be examined by the VT method in accordance with T9074-AS-GIB-010/271.

4.5.4.1.2 Criteria. Each forging shall be free of pipe, cracks, and flakes. Within the limits of good manufacturing practices, the forgings shall be free of injurious seams, laps, segregation, or other imperfections that, due to their nature, degree, or extent, will interfere with the use of the material in machining or fabrication of parts.

4.5.4.2 Magnetic particle testing (MT). MT shall be as specified in 4.5.4.2.1 and 4.5.4.2.2 (see 3.8.1).

4.5.4.2.1 Method. Each forging shall be examined by the MT inspection method in accordance with T9074-AS-GIB-010/271. For MT inspection of finished machined parts, the yoke method shall be used to prevent arc-strikes on the machined surface.

4.5.4.2.2 Criteria. All indications over ⅛ inch in length or any 6-square-inch area that exhibits 10 or more indications equal to or greater than 1/16 inch in length shall be rejected or repaired as specified in 3.8.

4.5.4.3 Ultrasonic testing (UT). When internal soundness is required (see 3.8.2), UT shall be as specified in 4.5.4.3.1 and 4.5.4.3.2 (see 3.8.2).

4.5.4.3.1 Method. UT shall be in accordance with T9074-AS-GIB-010/271. The forging shall be tested in both axial and longitudinal direction using either a shear or longitudinal wave, whichever is applicable to the geometry of the forging.

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4.5.4.3.2 Criteria. Unless otherwise specified (see 6.2), the forgings shall be free from all discontinuities that exceed those exhibited by the calibration standards in accordance with T9074-AS-GIB-010/271 and defects that cause complete loss of back reflection between parallel surfaces. Such defects shall be rejected or repaired as specified in 3.8.

4.6 Retests. Retests are permitted as specified in 4.6.1 and 4.6.2.

4.6.1 Chemical analysis retests. When a sample forging fails to meet the chemical requirements of table III, each forging in the lot may be analyzed to determine compliance with table III. The samples shall be taken as specified in 4.3.3.

4.6.2 Tension and impact retests. Retests shall be permitted only for the following listed conditions:

a. Elongation or reduction of area value is no greater than 2 percent below the minimum.

b. Ultimate tensile strength and yield strength are no greater than 1,000 pounds per square inch (lb/in^2) outside of the range limits.

c. The impact average is no greater than 2 ft-lbs below the minimum and no single test is greater than 5 ft-lbs below the minimum.

Retest specimens shall be taken adjacent to the locations where the original specimens were taken.

4.7 Reheat treatment. When the initial tension or impact tests, or the retests of 4.6.2, fail to meet the requirements of table IV or table V, reheat treatment of the lot shall be permitted. Reheat treatment shall be as specified in 3.5. After reheat treatment, the lot shall meet all the requirements of this specification.

4.8 Rejection. When any forging representing a lot fails to conform to the requirements of this specification, the lot shall be subjected to rejection, or each forging in that lot shall be tested and determined to meet the requirements of this specification.

4.9 Certificate of Conformance (CoC). Unless otherwise specified (see 6.2), a CoC shall be provided for each lot of material offered for acceptance (see 6.3). The CoC shall contain the following:

a. Quantitative data obtained from the required chemical and mechanical tests specified herein.

b. Qualitative results of nondestructive tests and other inspections.

c. Documentation of final heat treatment and stress relieving details, as applicable.

d. Documentation of the melting and casting processes.

e. Documentation when virgin raw material is used.

f. Documentation of the class, type, drawing or die number, heat, and heat treatment lot, and sizes that are marked on the forgings to provide traceability.

g. Documentation as to whether the material was forged or hot rolled.

h. Documentation of repairs and reference to approvals obtained, as applicable. If no repairs were required, the documentation shall indicate as such.

i. A statement that metal removal using flame cutting or other thermal means has not occurred after the forgings have received the final thermal treatment and final nondestructive tests.

j. A statement that each lot has been sampled, tested, and inspected in accordance with this specification and that the manufacturer has maintained manufacturing procedures and practices to produce forgings to meet the minimum property requirements throughout the forging.

k. A statement that each lot meets all requirements in accordance with this specification and shall be signed by a responsible representative of the contractor (see 6.6.1).

Where certificates issued by the manufacturer contain the above information, a separate certificate will not be requested from the contractor.

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4.10 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein. Packaging shall be inspected for compliance with the contract (see 5.1, 6.2, and 6.4).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The carbon and alloy steel forgings covered by this specification are intended for various shipboard applications. This specification contains alloys and requirements specific for use in military applications. Current stock of steel forgings may be used until depleted; future requisitions for steel forgings should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The class and type of steel forgings required (see 1.2).
c. The specific issue of individual documents referenced (see section 2).
d. Whether continuous casting is prohibited (see 3.1.1).
e. Whether special melting practices are required (see 3.1.1).
f. Whether virgin raw materials are required (see 3.1.4).
g. Alternate forging process requirements (see 3.2).
h. When transverse mechanical properties are required (see table IV, note 3).
i. When impact testing is required (see 3.4.2).
j. Any stress relief treatment required (see 3.5.3).
k. Dimensions and tolerances required (see 3.6).
l. When identification marking is not required (see 3.7).
m. When marking other than indentation stamping is required (see 3.7).
n. When internal soundness is required (see 3.8.2).
o. Details of the approval process required for forgings to be repaired by welding (see 3.8.4).
p. Requirements for repair by welding (e.g., repair limits, welding, fabrication, inspection, heat treatment, and stress relief) (see 3.8.4).
q. Ultrasonic examination acceptance criteria if other than as specified (see 4.5.4.3.2).
r. When a certificate of conformance (CoC) is not required (see 4.9).
s. Packaging requirements (see 5.1 and 6.4).

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6.3 CoC considerations. The CoC as required in this specification is an essential element for assuring that the material being purchased is in conformance with the specification. The certificate content, as required herein, provides valuable information regarding the test results, expected performance, actual weld metal chemistry, the specific heat treatment applied, and other information that links the certificate to the manufacturer, purchaser, specification, lot, etc. Navy technical and safety program documentation typically require certain information included in the prescribed CoC for use as objective quality evidence (OQE) demonstrating specification conformance and compliance as well as traceability for materials used within prescribed ship systems. Such documentation includes, but is not limited to, SUBSAFE, Level I, Deep Submergence Scope of Certification (DSSOC), and ship hull, machinery, and piping systems fabrication documents. A CoC may also be required by customs. The CoC also provides beneficial information in support of failure analyses, tracking problematic heats or lots of material throughout the supply system or onboard ships, and material, component, and specification improvements. These added benefits are possible because of the unique tracking information provided within the CoC and the unique set of mechanical properties and chemical content provided for each lot of material.

6.4 Packaging recommendations. Packaging recommendations are provided in 6.4.1 through 6.4.4.

6.4.1 Preservation. Preservation should be sufficient to protect against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early material use.

6.4.2 Packing. Packing should be accomplished in a manner that will ensure acceptance by a common carrier and will afford protection against physical damage during shipment from the supply source to the using activity for early material use.

6.4.3 Marking. Shipment marking information should be provided on interior packaged and exterior shipping containers in accordance with the contractor's commercial practice. The information should include nomenclature, national stock number, contract or order number, contractor's name, and the destination.

6.4.4 Commercial preservation, packing, and marking. Unless otherwise specified (see 6.2), the material should be packaged in accordance with ASTM A700.

6.5 Supersession data. This specification supersedes MIL-S-24093A(SH), dated 5 November 1991, MIL-S-24093A(SH) w/INT AMENDMENT 1, dated 10 July 2012, MIL-S-24093/1, dated 5 November 1991, MIL-S-24093/2, dated 5 November 1991, and MIL-S-24093/3, dated 5 November 1991.

6.6 Definitions. Definitions are provided in 6.6.1 through 6.6.5.

6.6.1 Contractor. The contractor is the seller under the contract or purchase order that incorporates this specification.

6.6.2 Environmentally preferable material. Products or services that have a lesser or reduced effect on human health and the environment when compared with competing products or services that serve the same purpose. This comparison may consider raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance or disposal of the product or service.

6.6.3 Manufacturer. The manufacturer is the producer that is responsible for the fabrication of the forging, as required by this specification.

6.6.4 Recovered material. Waste materials and by-products that have been recovered or diverted from solid waste. It does not include those materials and by-products generated from, and commonly reused within, an original manufacturing process.

6.6.5 Recycled material. Product or other material recovered from the solid waste stream for use in the form of raw materials in the manufacture of new products other than fuel for producing heat or power by combustion.

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6.7 PIN and cross-reference of classifications. Marking requirements are contained in 3.7. For some classes of steel forgings, marking includes a part numbering system. A definitive new part numbering system is formulated as described in figure 1. This new part numbering system has been established to incorporate the PINs from MIL-S-24093/1, MIL-S-24093/2, and MIL-S-24093/3 relative to specific round and hexagonal bars. Table VII provides for the old part numbers and is correlated to the new part numbers. The new part numbers should be used. The part numbering system is used only for the items shown in table VII. This specification, however, does not limit bars to these classes, types, sizes, or shapes. Figure 2 provides a depiction of the A dimension and length defined in table VII.

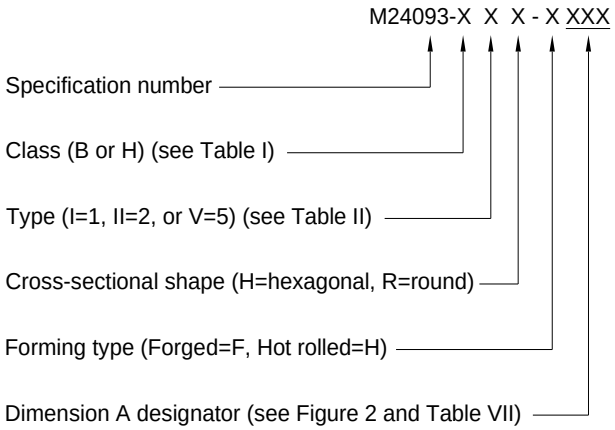


FIGURE 1. Part numbering system.

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TABLE VII. PIN cross-reference.

Previous Specification	Class	Type	Dim. A (in.)	Dim. A Designator	Dim. L (ft)	Cross-sectional Shape	Forging Process	Old PIN	New PIN
MIL-S-24093/1	B	I	1.250	001	10 to 12	Round	Forged	M24093/1-001	M24093-B1R-F001
MIL-S-24093/1	B	I	1.312	003	10 to 12	Round	Forged	M24093/1-003	M24093-B1R-F003
MIL-S-24093/1	B	I	1.500	005	10 to 12	Round	Forged	M24093/1-005	M24093-B1R-F005
MIL-S-24093/1	B	I	1.875	007	10 to 12	Round	Forged	M24093/1-007	M24093-B1R-F007
MIL-S-24093/1	B	I	2.705	009	10 to 12	Round	Forged	M24093/1-009	M24093-B1R-F009
MIL-S-24093/1	B	I	1.250	001	10 to 12	Round	Hot Rolled	M24093/1H001	M24093-B1R-H001
MIL-S-24093/1	B	I	1.312	003	10 to 12	Round	Hot Rolled	M24093/1H003	M24093-B1R-H003
MIL-S-24093/1	B	I	1.500	005	10 to 12	Round	Hot Rolled	M24093/1H005	M24093-B1R-H005
MIL-S-24093/1	B	I	1.875	007	10 to 12	Round	Hot Rolled	M24093/1H007	M24093-B1R-H007
MIL-S-24093/1	B	II	1.250	001	10 to 12	Round	Forged	M24093/1-001	M24093-B2R-F001
MIL-S-24093/1	B	II	1.312	003	10 to 12	Round	Forged	M24093/1-003	M24093-B2R-F003
MIL-S-24093/1	B	II	1.500	005	10 to 12	Round	Forged	M24093/1-005	M24093-B2R-F005
MIL-S-24093/1	B	II	1.875	007	10 to 12	Round	Forged	M24093/1-007	M24093-B2R-F007
MIL-S-24093/1	B	II	2.705	009	10 to 12	Round	Forged	M24093/1-009	M24093-B2R-F009
MIL-S-24093/1	B	II	1.250	001	10 to 12	Round	Hot Rolled	M24093/1H001	M24093-B2R-H001
MIL-S-24093/1	B	II	1.312	003	10 to 12	Round	Hot Rolled	M24093/1H003	M24093-B2R-H003
MIL-S-24093/1	B	II	1.500	005	10 to 12	Round	Hot Rolled	M24093/1H005	M24093-B2R-H005
MIL-S-24093/1	B	II	1.875	007	10 to 12	Round	Hot Rolled	M24093/1H007	M24093-B2R-H007
MIL-S-24093/2	H	V	4.125	001	10 to 12	Hexagonal	Forged	M24093/2-001	M24093-H5H-F001
MIL-S-24093/3	B	I	0.562	001	10 to 12	Hexagonal	Forged	M24093/3-001	M24093-B1H-F001
MIL-S-24093/3	B	I	0.875	003	10 to 12	Hexagonal	Forged	M24093/3-003	M24093-B1H-F003
MIL-S-24093/3	B	I	0.938	005	10 to 12	Hexagonal	Forged	M24093/3-005	M24093-B1H-F005
MIL-S-24093/3	B	I	1.291	007	10 to 12	Hexagonal	Forged	M24093/3-007	M24093-B1H-F007
MIL-S-24093/3	B	I	2.000	009	10 to 12	Hexagonal	Forged	M24093/3-009	M24093-B1H-F009
MIL-S-24093/3	B	I	2.250	011	10 to 12	Hexagonal	Forged	M24093/3-011	M24093-B1H-F011
MIL-S-24093/3	B	I	2.250	(see note 1/)	10 to 12	Hexagonal	Forged	M24093/3-013	(see note 1/)
MIL-S-24093/3	B	I	2.562	015	10 to 12	Hexagonal	Forged	M24093/3-015	M24093-B1H-F015
MIL-S-24093/3	B	I	2.750	017	10 to 12	Hexagonal	Forged	M24093/3-017	M24093-B1H-F017
MIL-S-24093/3	B	I	0.562	001	10 to 12	Hexagonal	Hot Rolled	M24093/3H001	M24093-B1H-H001
MIL-S-24093/3	B	I	0.875	003	10 to 12	Hexagonal	Hot Rolled	M24093/3H003	M24093-B1H-H003
MIL-S-24093/3	B	I	0.938	005	10 to 12	Hexagonal	Hot Rolled	M24093/3H005	M24093-B1H-H005
MIL-S-24093/3	B	I	1.291	007	10 to 12	Hexagonal	Hot Rolled	M24093/3H007	M24093-B1H-H007
MIL-S-24093/3	B	I	2.000	009	10 to 12	Hexagonal	Hot Rolled	M24093/3H009	M24093-B1H-H009

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TABLE VII. PIN cross-reference – Continued.

Previous Specification	Class	Type	Dim. A (in.)	Dim. A Designator	Dim. L (ft)	Cross-sectional Shape	Forging Process	Old PIN	New PIN
MIL-S-24093/3	B	II	0.562	001	10 to 12	Hexagonal	Forged	M24093/3-001	M24093-B2H-F001
MIL-S-24093/3	B	II	0.875	003	10 to 12	Hexagonal	Forged	M24093/3-003	M24093-B2H-F003
MIL-S-24093/3	B	II	0.938	005	10 to 12	Hexagonal	Forged	M24093/3-005	M24093-B2H-F005
MIL-S-24093/3	B	II	1.291	007	10 to 12	Hexagonal	Forged	M24093/3-007	M24093-B2H-F007
MIL-S-24093/3	B	II	2.000	009	10 to 12	Hexagonal	Forged	M24093/3-009	M24093-B2H-F009
MIL-S-24093/3	B	II	2.250	011	10 to 12	Hexagonal	Forged	M24093/3-011	M24093-B2H-F011
MIL-S-24093/3	B	II	2.250	(see note 1/)	10 to 12	Hexagonal	Forged	M24093/3-013	(see note 1/)
MIL-S-24093/3	B	II	2.562	015	10 to 12	Hexagonal	Forged	M24093/3-015	M24093-B2H-F015
MIL-S-24093/3	B	II	2.750	017	10 to 12	Hexagonal	Forged	M24093/3-017	M24093-B2H-F017
MIL-S-24093/3	B	II	0.562	001	10 to 12	Hexagonal	Hot Rolled	M24093/3H001	M24093-B2H-H001
MIL-S-24093/3	B	II	0.875	003	10 to 12	Hexagonal	Hot Rolled	M24093/3H003	M24093-B2H-H003
MIL-S-24093/3	B	II	0.938	005	10 to 12	Hexagonal	Hot Rolled	M24093/3H005	M24093-B2H-H005
MIL-S-24093/3	B	II	1.291	007	10 to 12	Hexagonal	Hot Rolled	M24093/3H007	M24093-B2H-H007
MIL-S-24093/3	B	II	2.000	009	10 to 12	Hexagonal	Hot Rolled	M24093/3H009	M24093-B2H-H009

NOTE:

1/ The old PIN M24093/3-013 for Class B, represented types I and II and an "A" dimension of 2.250. The old PIN appeared as a duplicate of PIN M24093/3-011 and, therefore, PIN M24093/3-013 is no longer represented with a new PIN.

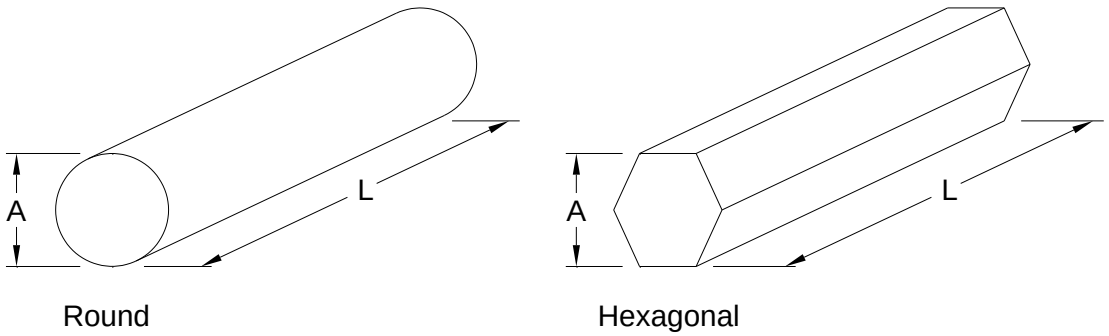


FIGURE 2. Dimensions of round and hexagonal bars conforming to the new PINs.

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6.8 Subject term (key word) listing.

Casting

Ingot, as-cast

Testing, impact

Testing, tension

Tests, nondestructive

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project FORG-2024-001)

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